**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 094 · Lesson 094 · CPU Performance: Clock, Cores and Cache

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a cache hit?

Q2. What is a cycle?

Q3. What is a core?

Q4. A search space of 64 candidates is repeatedly halved to one. How many halvings?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

GHz — Billions of cycles per second.

core — CPU execution core.

cache hit — Data found in CPU cache.

IPC — Instructions retired per cycle (workload dependent).

## New learning

Observe — Clock speed counts clock cycles per second; one cycle is not always one complete instruction. Frequency alone cannot predict program performance.

Explain — Multiple CPU cores improve performance most for work divided into independently executable threads; a strictly sequential task may gain little.

Evaluate — Cache is small, fast storage close to CPU. A cache hit avoids some slower RAM accesses; cache size, locality and hit rate influence results.

## Worked example

Compare 2 GHz and 3 GHz clock rates under equal work per clock and no bottlenecks.

1. Explain the mechanism — 3 GHz offers 50% more cycles each second than2 GHz; this does not universally imply 50% faster program runtime. — The comparison requires equal microarchitecture and workload assumptions.

# We Do · Practise with guidance

A single-threaded algorithm runs on a 2-core or 8-core CPU of equal per-core performance. Explain likely benefit.

Explain the mechanism

It may not be substantially faster from extra cores alone; only one thread is the bottleneck.

Parallel benefits need tasks/threads that can run concurrently.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain effects of clock speed, number of cores and cache on performance. [6]

### Question 2

CPU X 2GHz and Y 3GHz have identical IPC; compute ratio of cycles/sec and one reason runtime may differ from ratio. [6]

### Question 3

Recommend improvements for school video rendering versus a sequential exam-marking script. Discuss CPU performance factors. [6]

## Exit Ticket · Answer before checking

A 4GHz processor always completes any program twice as fast as a 2GHz processor.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Requested item found in cache.

Q2. A unit of processor clock timing.

Q3. An independent CPU execution unit.

Q4. 64→32→16→8→4→2→1; six halvings.

### You Do pages · Independent questions

Q1. Clock affects cycles/sec [2], cores parallel tasks [2], cache reduces slower memory accesses [2].

Q2. 3/2=1.5 [2], Y50% more cycles/s [2], memory stalls/parallelisation or instruction mix [2].

Q3. Rendering often exploits cores [2], sequential script needs stronger single-thread performance [2], cache/IPC/workload caveats [2].

### Exit Ticket · Model answer

False. Clock speed alone does not determine total performance.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**